8} Oxytocin Administration during First Stage

Goal

To enable nursing students to understand the principles and safe practices of oxytocin administration during the first stage of labor, promoting effective labor progression while safeguarding maternal and fetal health

Learning objectives: -

At the end of this skill, students will be able to: -

1- Define the action of oxytocin.
2- List the indication and contraindication of oxytocin administration.
3- Recognize the complication of oxytocin administration.
4- Identify the methods of oxytocin administration.
5- Describe the preparation of the equipment used.
6- Demonstrate the procedure of administering oxytocin during labor.

Outline:-

1. Introduction.
2. Action of oxytocin.
3. Indication of oxytocin.
4. Contraindication of oxytocin.
5. Complication of oxytocin.
6. Methods of administration.
7. Equipment

8. The procedure of administering oxytocin during labor.

Introduction:-

Oxytocin is a natural hormone that manages key aspects of female reproductive system, including labor and delivery and lactation. During labor it stimulates the uterus to contract, thereby facilitating cervical dilatation. The doctor may use synthetic forms

of oxytocin (Pitocin or syntocinon) to induce or augment labor if it hasn't started naturally or strengthen uterine contractions and to control bleeding after the placenta is delivered.

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Oxytocin

Oxytocin is a neuropeptide produced in the hypothalamus and secreted into the bloodstream thought the posterior lope of pituitary gland where they interact with oxytocin receptors throughout the body.

Syntocinon (Pitocin):

It is a synthetic form of oxytocin which acts as a uterine stimulant.

Pharmacological Action of Oxytocin

Synthetic oxytocin acts specifically on the pregnant uterus. It increases the frequency, strength, and duration of uterine contractions. Its effects begin within minutes and have a short half-life of 3–5 minutes, allowing for rapid titration. It also has an antidiuretic effect, especially at high doses, which can lead to water intoxication if not properly monitored

Indication of oxytocin administration:-

1- Management of certain type of abortion as inevitable, incomplete and missed abortion to help the uterus to expel its contents.
2- Post-abortive bleeding.
3- During evacuation of hydatiform mole to avoid perforation of uterus and excessive bleeding.
4- Induction of labor.
5- Oxytocin stress test to assess fetal wellbeing.
6- Uterine inertia during labor
7- Abruption placenta.
8- In the management of 3rd stage of labor.
9- Postpartum hemorrhage (PPH) management
10- During cesarean section.
11- Breast engorgement: oxytocin nasal drops or spray at time of breast feeding to increase milk flow.

Contraindication of syntocinon administration:-

- Obstructed labor.

- Placental insufficiency.
- Malpresentation as transverse or oblique lie.
- Scare in the uterus (relative contraindication).
- Grand multipara.
- Fetal distress.

Complications of syntocinon administration:-

1- Rupture of the uterus, cervical laceration, contraction ring.

2- Secondary uterine inertia with subsequent PPH.

3- Fetal asphyxia, amniotic fluid embolism, coronary spasm and shock, water intoxication if a large dose is given because of antidiuretic effect.

4- Neonatal hyperbilirubinemia and jaundice.

Methods of administration:-

1- **Orally: -**A tablet is placed between the gum and cheek and left to dissolve slowly in 30 minutes. (It can be repeated every ½ hour).

2- **Intramuscularly:** but this is avoided before delivery of the fetus.

3- **Intrauterine:** - during cesarean section.

4- **Intravenously:** - by the drip method and this is the best.

5- **Nasal spray or drops.**

Recent Clinical Updates and Evidence-Based Recommendations

- Prefer low-dose oxytocin protocols to reduce risk of uterine hyperstimulation.
- Use oxytocin cautiously in VBAC cases due to risk of uterine rupture.

- Incorporate smart infusion pumps with safety limits to minimize human error.
- Individualize postpartum oxytocin dose to uterine tone and bleeding risk.
- Counsel women about oxytocin use to ensure informed consent and engagement.
- Monitor for signs of water intoxication in high-dose or long-duration

Equipments:-

- Intravenous cannula. - Alcohol swabs - Elastic adhesive tab
- Torniquet. - IV set - I.v solution (primary line)
- Oxytocin solution - Syringe with normal saline - IV stand
- External or internal fetal monitoring - Fluid balance chart - Label

Preparation of IV solution:-

1. Prepare the oxytocin solution as doctor prescribed.
2. Rotate the I.V bottle to dissolve the drug throughout the solution.
3. Label the I.V bottle with the medication name and dose.

8} Procedure of Administering Oxytocin during Labor

Action	Rational
1-Great the mother, introduce your and explain procedure to the mother.	- To gain cooperation, helps minimize anxiety during procedure
2- Prepare equipment & take it to bed side	
3- Provide privacy.	
4- Wash hands.	
6- Help the mother to assume the lateral position, or dorsal recumbent position. ☞ Don't let her lie supine.	- In the supine position, the gravid uterus presses on the maternal great vessels, producing maternal hypotension and reduced uterine perfusion.
6-Check the mother vital signs, position of fetus, uterine contraction, cervical dilatation and fetal heart rate before starting the infusion	-To get baseline data and identifying any deviation from these data.

7- Connect the woman with fetal monitor to assess fetal wellbeing and to establish a baseline for uterine activity	-A minimum of 20 minutes of fetal monitoring before starting the infusions is essential to establish the fetal heart rate baseline. -Don't start the oxytocin infusion if none reassuring fetal heart rate patterns are identified
8-Prepare the I.V solutions: Dilute the ampoule of oxytocin (10 IU) with 10 cm of ringer lactate or dextrose 5%. So, each cm of diluted solution contains 1 IU of oxytocin. ☞ Insert 5 cm of the prepared dose of oxytocin in the 500 cc I.V. fluid bottle. ☞ Make sure that the oxytocin solution is labeled. ☞ Connect the I.V. set in the bottle and hang it on the I.V. line stand to justify the I.V. set through the	- Start this line to deliver not only oxytocin but also fluid, blood, or other medications as needed.

infusion pump.	
9-Insert the IV cannula in the right hand and fix it with suitable size of plaster (using cotton with alcohol before insertion to clean the site of insertion): • Connect the prepared I.V. fluid with the inserted cannula. • Be sure that the right hand in a straight position • Support the woman's hand with pillow under her arm.	- Changes in arm position may alter the flow rate.

10- Turn on the oxytocin infusion at prescribed rate.	- The initial dose of oxytocin is usually 0.5 to 1.0 IU /minute at a rate of 10drops/minute
11- Assess and record the woman's blood pressure, uterine contractions (frequency, duration, intensity and interval) and the fetal heart rate before each increase in oxytocin dosage to evaluate the effect of oxytocin and determine the need to increase or stop the infusion.	- Because oxytocin beings acting immediately.
12- Increase the infusion rate by 10 drops per minute every 30 minutes until a good contraction pattern is established (contraction lasting more than 40 seconds and occurring three times in 10 minutes).	-Before each increase, be sure about the time the frequency, interval and duration of contractions.

13- Maintain this rate until delivery is completed to avoid uterine hyper stimulation	**Signs of uterine hyper stimulation** ☞ **Frequency-** : **more than 5** time/10 min ☞ **Duration:** -longer than 90 sec ☞ With or **without fetal heart** rate changes ☞ No **Interval** or less than 1 mint ☞ Contraction is **shallow** ☞ Not effect of **cervix**
14- Discontinue the oxytocin infusion and notify a physician if maternal or fetal complications have been developed such as: • Uterine hyperstimulation. • None reassuring fetal heart rate patterns.	
15- **Apply the following measures in case of hyperstimulation (overdosing):** ☞ Turn off oxytocin infusion.	

☞ Increasing the rate of primary infusion ☞ Positioning the woman on her left side. ☞ O2 by mask as ordered. ☞ Notify the doctor. ☞ Administering a tocolytic agent such as terbutalin to stop excessive uterine activity.	- To increase the uterine blood flow
16- Record fluid intake and output every 2 hours	-Output should be at least 30 ml/h
17- Collect equipment and clean it.	
18- Wash hands.	
19- Complete the partogram data.	
20- Let the oxytocin infusion to be completed after delivery with close	
21- Observation of any signs of postpartum hemorrhage until the dose that is finished.	

Oxytocin Administration Checklist

Action	First Trial		Second Trial	
	Done	Note Done	Done	Note Done
1- Greet the mother, introduce yourself and explain procedure to the mother.				
2- Prepare equipment & take it to bed side				
3- Provide privacy.				
4- Wash hands.				
5- Help the mother to assume the lateral position, or dorsal recumbent position.				
6- Identify and record the fetal heart rates, and assess uterine contractions occurring in a 10 minutes span.				
7- Insert the intravenous cannula.				
8- Connect the electrolytes I.V solution (primary line)				
9- Add the oxytocin solution to the primary I.V line at the Y injection site closest to the mother.				
10- Turn on the oxytocin infusion as prescribed rate.				

11- Monitor the uterine contraction immediately then monitor F.H.R., uterine contraction (frequency, duration, strength and interval) every 30 minutes.				
12- Increase the oxytocin dosage as prescription.				
13- When the uterine contraction occurring every 2 to 3 minutes and lasting 40 to 60 seconds you can stop increases in the oxytocin rate.				
14- Reposition the woman at her other side as needed.				
15- If hyper stimulation of the uterus occurs and fetal heart rate changes. The following action are taken: ☞ Turn off oxytocin infusion. ☞ Speed up primary infusion. ☞ Change position to the other side. ☞ Give mask oxygen. ☞ Notify the doctor.				
16- Record fluid intake and output every 2 hours				
17- Collect equipment and clean it.				
18- Wash hands.				
Procedure score				
Date				

Student name: **Instructor name**